# EmbMerge: A Transformer-based Method for Fusing CDR Lists

Mehmet Erdeniz Aydoğdu, Yağmur Duru Tüfekçioğlu, Ismail Sengor Altingovde,
Pinar Karagoz, and Ismail Hakki Toroslu • Middle East Technical University, Turkiye

## Introduction

- **The Problem**: Modern recommendation systems suffer from data sparsity and cold-start problems.
- **The Cross-Domain (CDR) Solution:** CDR methods transfer knowledge from a data-rich source domain to a sparse target domain.
- **The Challenge:** Different CDR models have varying architectures and capture unique, complementary signals. No single model perfectly captures complete user preferences.
- **Limitation of Existing Fusion:** Traditional rank fusion techniques rely on ranks or sparse, hand-crafted statistical features that fail to capture deep semantic relationships between items.

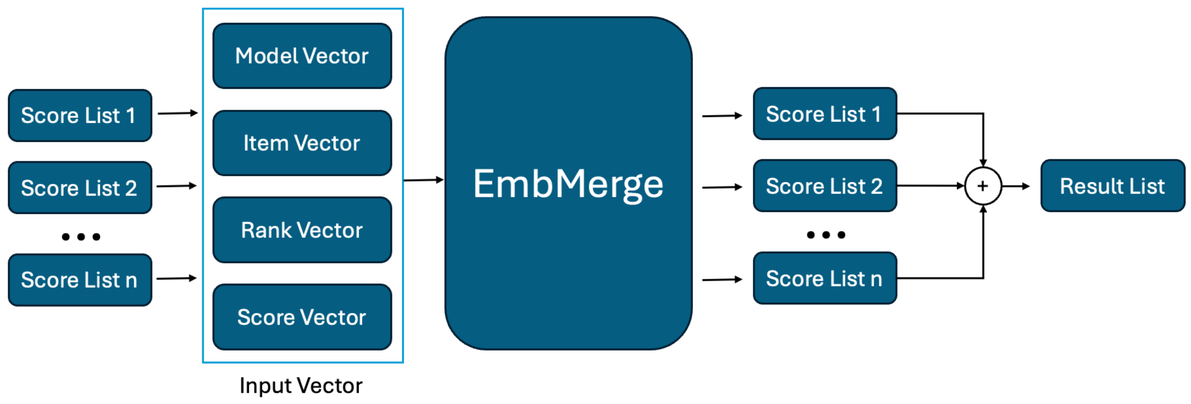
## EmbMerge

A supervised learning-to-rank method using dense embeddings and a Transformer encoder to fuse multiple CDR lists into a single, superior ranking. We propose two different variants: Standard and Direct Prediction Models.

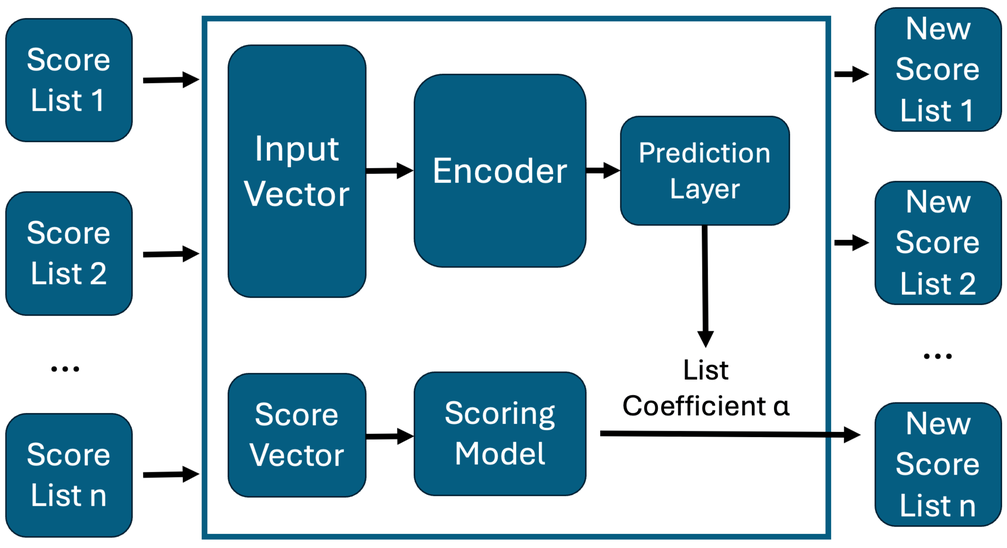
**Input Representation**
Instead of sparse features, we concatenate four dense vectors per item:

- **Model Vector**: Source CDR model.
- **Item Vector**: Unique item characteristics.
- **Rank Vector**: Position in the ranked list (e.g., 1–100).
- **Score Vector**: Normalized raw score.

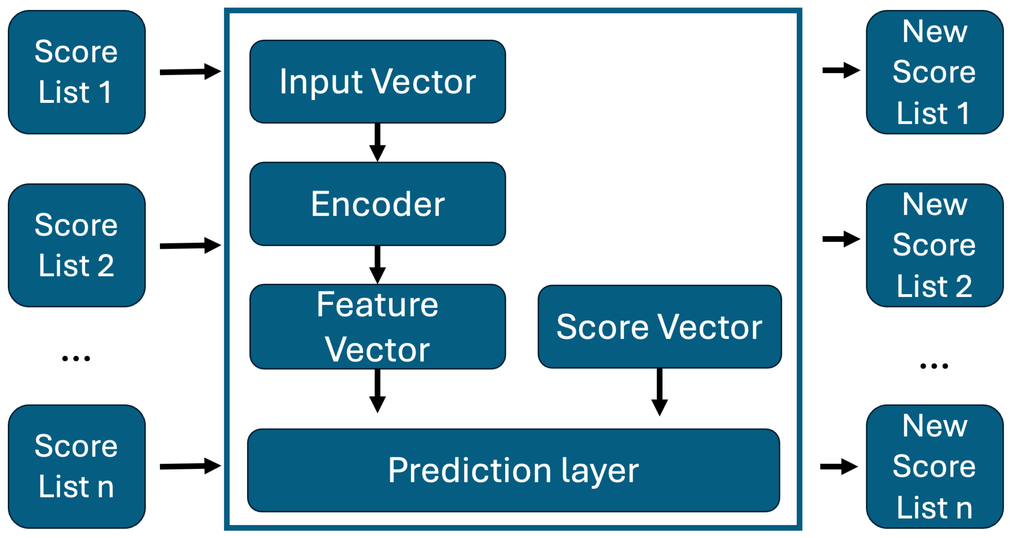
## Architecture




## Standard Model




## Direct Prediction Model




## Experiments

**Datasets:**

- **Digix:** Source: News Feeds → Target: Ads (439k test lists)
- **Amazon:** Source: Books → Target: Movies & TV Shows (130k test lists)

**Base CDR Models**: BiTGCF, CMF, and DTCDR. (1 positive and 99 negative items per list)

**Baselines for Comparison**: Unsupervised (RRF, BordaFuse) and Supervised (Logn-ISR, W-BordaFuse)

| | Digix | | Amazon | |
|---|---|---|---|---|
| **Methods / Metrics** | **Recall@10** | **NDCG@10** | **Recall@10** | **NDCG@10** |
| **Base CDR Models** | | | | |
| BiTGCF | 0.8904 | 0.7129 | 0.6242 | 0.385 |
| CMF | 0.8613 | 0.6729 | 0.6016 | 0.3853 |
| DTCDR | 0.878 | 0.6483 | 0.5739 | 0.3423 |
| **Fusion Baselines** | | | | |
| W-BordaFuse | 0.9038 | 0.7245 | 0.6541 | 0.4134 |
| BordaFuse | 0.9011 | 0.7191 | 0.6519 | 0.4095 |
| Logn-ISR | 0.8953 | 0.7086 | 0.6469 | 0.4062 |
| RRF | 0.9026 | 0.7187 | 0.6595 | 0.4145 |
| **EmbMerge (Ours)** | | | | |
| Standard Model | 0.9007 | 0.7237 | **0.6619** | **0.4215** |
| Direct Prediction | **0.9196** | **0.7279** | 0.6459 | 0.4035 |

## Conclusion

Either one of the EmbMerge variants consistently outperforms individual CDR models and strong supervised/unsupervised fusion baselines across different datasets.

Replacing sparse statistical features with learned, dense embeddings successfully captures the complex, deep-semantic relationships between models, items, ranks and scores.

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erdeniz.aydogdu@metu.edu.tr, yagmur.tufekcioglu@metu.edu.tr, altingovde@ceng.metu.edu.tr, karagoz@ceng.metu.edu.tr, toroslu@ceng.metu.edu.tr